Mehak Chopra

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EDUCATION

University of Galway — **Galway, Ireland**
Ph.D. Genomic Data Science — *September 2022–Present*

Pondicherry University — **Puducherry, India**
M.Sc. Bioinformatics — *July 2019–July 2021*

Panjab University — **Chandigarh, India**
B.Sc. Bioinformatics — *July 2016–May 2019*

RESEARCH EXPERIENCE

University of Galway — **Galway, Ireland**
PhD Research Candidate — *September 2022–August 2026*

PhD research focuses on uncovering the genomic and cellular factors influencing aortic distensibility and aneurysm development.

- Chapter 1 investigates the genomic basis of aortic distensibility by evaluating how inferred cellular composition and cell-type-specific gene expression contribute to age- and genotype-related variations.
- Chapter 2 assesses whether genes associated with distensibility exhibit altered expression, immune cell composition, and cell type–specific gene set analysis in blood from individuals with aortic aneurysms compared to healthy controls.
- Chapter 3 utilizes single-nucleus transcriptomics to characterize biomolecular differences between aneurysmal tissues of bicuspid and tricuspid aortic valve patients.

University of Liverpool — **Liverpool, UK**
Visiting Research Fellow — *March 2025*

- Gained practical experience working with biobanked aortic tissue samples from diseased vessels, involving exposure to collection procedures and an understanding of how biomechanical properties of aortic tissues are measured and analyzed for research applications.

Yale School of Medicine, Yale University — **New Haven, USA**
Postgraduate Fellow — *October 2024–January 2025*

- Collaborated closely with clinicians to deepen understanding of cardiovascular diseases while applying advanced statistical and bioinformatic methods to analyze gene expression and whole-exome sequencing data, leading to the identification of key genomic insights.

Centre for Brain Research, Indian Institute of Science — **Bangalore, India**
Data Analyst — *October 2021–August 2022*

- Managed and analyzed genotypic data from the Indian population as part of the Genome India Project, developing a copy number variation detection pipeline and conducting evolutionary genomics analyses to trace gene flow and evolutionary links between ancient and modern human populations.

Centre for Bioinformatics, Pondicherry University — **Puducherry, India**
MSc Project — *February 2021–June 2021*

- Undertook a Master's research project titled "Triad of Coronary Heart Disease, Hyperlipidemia, and Diabetes", aiming to identify shared molecular signatures among these conditions through differential gene expression analysis of RNA-seq data, complemented by gene ontology, enrichment, and pathway analyses.

GGDSD College, Panjab University **Chandigarh, India**
BSc Project *January 2018–April 2018*

- Undertook a Bachelor's project titled "In Silico Mutational Analysis of Interleukin-10", which involved using bioinformatics approaches to induce mutations in protein structures and assess the stability differences between mutant and wild-type proteins through receptor docking analysis.

PUBLICATIONS

- **Chopra M**, Hynes N, Seoighe C. Impact of cell type specific variations and age in aortic distensibility. medRxiv. 2025 Nov 27:2025-11. doi: 10.1101/2025.11.25.25340959. *(preprint)*
- Seoighe C, Connaire S, **Chopra M**. Probing the limits of cis-acting gene regulation using a model of allelic imbalance quantitative trait loci. PLoS genetics. 2025 Apr 30;21(4):e1011446. doi: 10.1371/journal.pgen.1011446.
- Subramanian K, **Chopra M**, Kahali B. Landscape of genomic structural variations in Indian population-based cohorts: deeper insights into their prevalence and clinical relevance. Human Genetics and Genomics Advances. 2024 Jul 18;5(3). doi: 10.1016/j.xhgg.2024.100285.
- Kaur A, **Chopra M**, Bhushan M, Gupta S, Kumari P H, Sivagurunathan N, Shukla N, Rajagopal S, Bhalothia P, Sharma P, Naravula J. The omic insights on unfolding saga of COVID-19. Frontiers in immunology. 2021 Oct 20;12:724914. doi: 10.3389/fimmu.2021.724914.

TEACHING EXPERIENCE

Teaching Assistant **University of Galway**
M.Sc. in Genomic Data Science *September 2023 - December 2025*

As a teaching assistant, I prepared and presented material for practical sessions, led training sessions, and provided feedback to the teaching team based on submission grades for the following modules:

- MA5116 - Introduction to Probability for Genomics
- MA215 - Mathematical Molecular Biology I
- MA216 - Mathematical Molecular Biology II
- MA5106 - Medical Genomics I

GRANTS

Open Targets Hackathon Funding **EMBL**
Lead Applicant *October 2025*

- Secured sponsorship to attend the Open Targets Hackathon; collaborated on a project that received the Best Project Award.

Research Mobility Grant **European Ageing Network**
Lead Applicant *March 2025*

- Funded a one-month research visit to the University of Liverpool to collaborate with the LABB (Liverpool Aortic Biomechanics and Biochemistry Research) group; gained hands-on experience in assessing aortic biomechanical properties.

ASFHG African Genomics Proposal **CRT Genomics data science**
Group Application *February 2023*

- Received full travel and accommodation support from CRT for presenting a selected research proposal on African population genomics at the ASFHG conference in Cape Town, South Africa.

RESEARCH COMPLIANCE

April, 2024: Prepared and submitted ethics application for a study on molecular and biomechanical

properties of healthy and diseased aortic tissue; successfully obtained approval from the HSE Galway Ethics Research Committee, enabling sample collection and pilot study execution.

February, 2025: Organized and conducted mini-competition among vendors in line with University of Galway's procurement regulations, ensuring transparency and compliance.

SELECTED PRESENTATIONS

- Presented ongoing PhD research on the genomic and cellular determinants of aortic distensibility in multiple national and international meetings, through both oral and poster formats: *Surgical Research Society Annual Meeting 2026 (London, UK); VIBE 2025 (Dublin, Ireland); EuroAgeNet "Cell Communication Networks in Ageing and Disease" Conference 2025 (Liverpool, Uk); Aorta Meeting 2024 (Liverpool, UK); Indian Society of Human Genetics 2024 (ISHG'24) (Gujarat, India); Irish Computational Biology and Genomics Symposium 2023 (ICBG'23) (Galway, Ireland); and School of Mathematical and Statistical Sciences Research Day 2025 (University of Galway, Ireland).*
- Recognitions include *Best Poster (EuroAgeNet 2025)*, *Best Lightning Talk (School of Mathematics and Statistics Research Day 2025)*, *Student Scholar Award (ISHG'24)*, and *Best Presentation (ICBG'23)*.

POSITIONS OF RESPONSIBILITY

- One of two student representatives for my PhD cohort (2022–2026), which involved planning events and liaising with the program managers if any students had issues.
- Served on the PhD Programme Governance Committee (2023–2026), contributing student perspectives to institutional decisions and policy discussions.
- Mentored an undergraduate student on a research project (2024), providing guidance in data analysis.
- Member of the Organising Committee, Virtual Institute of Bioinformatics and Evolution/Irish Computational Biology and Genomics (VIBE/ICBG) Symposium (2023 and 2024).

SKILLS

- Skilled in NGS data analysis, image segmentation, and integration of molecular and imaging modalities.
- Extensive experience in analysing GWAS, WES, bulk and single-cell/nuclei RNA-seq datasets using advanced statistical and machine learning approaches.
- Proficient in R and Python for data processing, visualisation, and reproducible workflow development.
- Experienced with HPC, cloud computing, containerisation (Docker, Singularity), and workflow management (Nextflow).
- Strong understanding of cardiovascular biology, particularly aortic and vascular genomics.
- Effective scientific communicator with a collaborative, literature-driven research approach.

REFERENCES

Prof. Cathal Seoighe
Professor of Bioinformatics and Scientific Director
Research Ireland Centre for Research Training in Genomics Data Science
University of Galway
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Dr. Niamh Hynes
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Department of Vascular Surgery and Endovascular Surgery
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